

Village-Level Thresholds

One Earth Balance Sheet — Malaysia 2026

Purpose: Centralized reference for all thresholds used in village-level resource scoring and balance sheet calculations.

Source: Research-based standards from WHO, UN-Habitat, Malaysian government, and ecological guidelines.

Domain 1: Natural Resources & Environment

1.1 Land Thresholds

Metric	Threshold	Source & Link	Notes
Population Density	400-2000 pp/km	Rural Settlement Standards (http://unhabitat.org/sites/default/files/download-manager-files/A%20New%20Strategy%20of%20Sustainable%20Neighbourhood%20Planning%20Five%20principles.pdf)	< 400 = underpopulated, > 5000 = overcrowded
Forest Cover	60% of total area	Bhutan Constitutional Mandate (Article 5) (https://www.constituteproject.org/constitution/Bhutan_2008)	Critical for ecological resilience
Land Utilization	No fixed threshold	Context-dependent	Ratio of utilized land to total owned

Scoring Ranges:

- Density: 400-2000/km = 1.0, 2000-5000/km = 1.00.5 (linear), >5000/km = 0.50.0 (linear)
- Forest: 60% = 1.0, 40-60% = 0.51.0 (linear), <40% = 0.00.5 (linear)

1.2 Water Thresholds

Metric	Threshold	Source & Link	Notes
Water Consumption	50-100 L/person/day	WHO Domestic Water Guidelines (https://www.who.int/water_sanitation_health/diseases/WSH03.02.pdf)	Includes drinking, cooking, and sanitation
Water Quality	Potable / Filtered / Contaminated	WHO Water Quality Standards (https://www.who.int/publications/i/item/9789241549950)	Potable = 1.0, Filtration = 0.5, Contaminated = 0.0
Scarcity Rate	<20% of households	Project Sustainability Standard	>50% households reporting scarcity = critical

Scoring Ranges:

- Scarcity: 0% = 1.0, <20% = 0.7, 20-50% = 0.4, >50% = 0.0
- Quality: Use household-level Q0315 scores, aggregate to village mean

1.3 Forestry & Timber Thresholds

Metric	Threshold	Source & Link	Notes
Forest Cover	60% minimum	Bhutan Constitutional Mandate (https://www.worldwildlife.org/projects/bhutan-committed-to-conservation)	Reused across domains
Sustainable Harvest	Harvest Regeneration	Biological Principle	Compute surplus/deficit
Forest Condition	Healthy / Disturbed / Degraded	Project Health Assessment	Healthy = 1.0, Disturbed = 0.6, Degraded = 0.3, Diseased = 0.0-0.2

1.4 Energy & Fuel Thresholds

Metric	Threshold	Source & Link	Notes
Grid Connectivity	100% target	Infrastructure Goal	Or equivalent local renewable self-sufficiency
Renewable Energy	50% of total consumption	SDG 7 Target Framework (https://sdgs.un.org/goals/goal7)	80%+ = excellent
Renewable Diversity	3+ types	Energy Resilience Goal	1-2 = moderate, 0 = poor

Scoring Ranges:

- Renewable %: 80% = 1.0, 50-80% = 0.7, 20-50% = 0.4, <20% = 0.1

Domain 2: Human Settlements & Infrastructure

2.1 Housing Thresholds

Metric	Threshold	Source & Link	Notes
Space per Person	15-25 m	UN-Habitat Adequate Housing (https://unhabitat.org/sites/default/files/2022/10/adequate-housing-and-living-conditions-eng.pdf)	>25m = excellent, <15m = inadequate
Housing Surplus	>10% extra units	Village Planning Target	-10% to +10% = balanced, <-10% = deficit
Construction Materials	Organic/Eco-friendly preferred	Ecological Building Specs	Traditional/Eco = 1.0, Wooden = 0.7, Concrete = 0.5

Scoring Ranges:

- Space: 25 m = 1.0, 15-25 m = 0.51.0 (linear), <15 m = 0.00.5 (linear)
- Surplus: >10% = 1.0, -10% to +10% = 0.7, <-10% = 0.70.0 (linear decline)

2.2 Community Infrastructure Thresholds

[!CAUTION]

NO APP DATA AVAILABLE - These thresholds are defined for future data collection

Facility Type	Threshold	Source & Link	Notes
Early Childhood Center	1 per 500 population	Default Planning Standard	Minimum Access Target
Primary School	1 per 1000 population	Default Planning Standard	Minimum Access Target
Healthcare Clinic	1 per 2000 population	WHO Healthcare Access (https://www.who.int/)	Minimum Access Target
Sewage Coverage	100% target	Sanitation Standard	Minimum Access Target

2.3 Mobility & Transportation Thresholds

Metric	Threshold	Source & Link	Notes
Road Connectivity	Paved / Unpaved / None	Infrastructure Target	Paved = 1.0, Unpaved = 0.5, None = 0.0
Unmet Mobility Needs	<20% households	Adequacy Threshold	>50% households reporting unmet need = critical

Domain 3: Livelihoods & Wellbeing

3.1 Agriculture / Food Thresholds

Metric	Threshold	Source & Link	Notes
Food Self-Sufficiency	90% within 100 miles	Localized Food System Target	<50% = poor
Soil Health (Stable)	80% of agricultural land	Ag-Sustainability Standard	Stable = 1.0, Disturbed = 0.6, Degraded = 0.3
Dietary Diversity	9 food categories	Nutritional Adequacy Target	Out of 12 predefined categories. 6-8 = Good

Scoring Ranges:

- Self-Sufficiency: 90% = 1.0, 70-90% = 0.71.0 (linear), 50-70% = 0.50.7 (linear), <50% = 0.00.5 (linear)
- Dietary Diversity: 9 = 1.0, 6-8 = 0.7, 3-5 = 0.4, <3 = 0.1

3.2 Business & Commercial Thresholds

Metric	Threshold	Source & Link	Notes
Business Activity	30% households	Economic Diversification	Excluding basic farming. <10% = Low
Business Profitability	80% businesses profitable	Economic Health	40-60% = Concerning. <40% = Poor

Scoring Ranges:

- Activity: 30% = 1.0, 20-30% = 0.7, 10-20% = 0.4, <10% = 0.00.4 (linear)
- Profitability: 80% = 1.0, 60-80% = 0.7, 40-60% = 0.4, <40% = 0.00.4 (linear)

3.3 Income & Expenditure Thresholds

Metric	Threshold	Source & Link	Notes
Household Surplus	80% households in surplus	Financial Health	Income > Expenditure

Scoring Ranges:

- Surplus: 80% = 1.0, 60-80% = 0.7, 40-60% = 0.4, <40% = 0.00.4 (linear)

3.4 Personal & Household Items Thresholds

Metric	Threshold	Source & Link	Notes
Local Production	80% of 6 categories	Self-Sufficiency Goal	Care, cleaning, office, medicine, kitchen, other

Standard Surplus/Deficit Score Formula

For any resource where available versus. needed is computed:

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ratio = available / needed (if needed > 0, else 1.0)

if ratio >= 1.2: # 20%+ surplus

score = 1.0

elif ratio >= 1.0: # Balanced or small surplus

score = 0.8

elif ratio >= 0.8: # Small deficit (within 20%)

score = 0.5

elif ratio >= 0.5: # Significant deficit (20-50%)

score = 0.3

else: # Severe deficit (>50%)

score = 0.0

Applied to: Land, Water supply, Housing units, Food, Energy, Timber, Mobility

Notes for Malaysia 2026 Deployment

1. **Contextual Adjustment:** Some thresholds may need adjustment based on:

- Specific bio-region (coastal versus. inland versus. mountainous)
- Local government standards
- Cultural norms (e.g., housing materials, space expectations)

1. **Periodic Review:** Thresholds should be reviewed and updated:

- After first data collection cycle (validate assumptions)
- Annually based on Malaysian policy updates
- When expanding to new bio-regions

1. **Data-Driven Calibration:** After collecting data from multiple villages, use statistical analysis to:

- Validate threshold appropriateness
- Identify outliers
- Adjust scoring bands if needed

Last Updated: February 13, 2026 | By Yagyaansh

Status: Ready for Malaysia 2026 pilot deployment